



UPPSALA
UNIVERSITET

Informationsteknologi

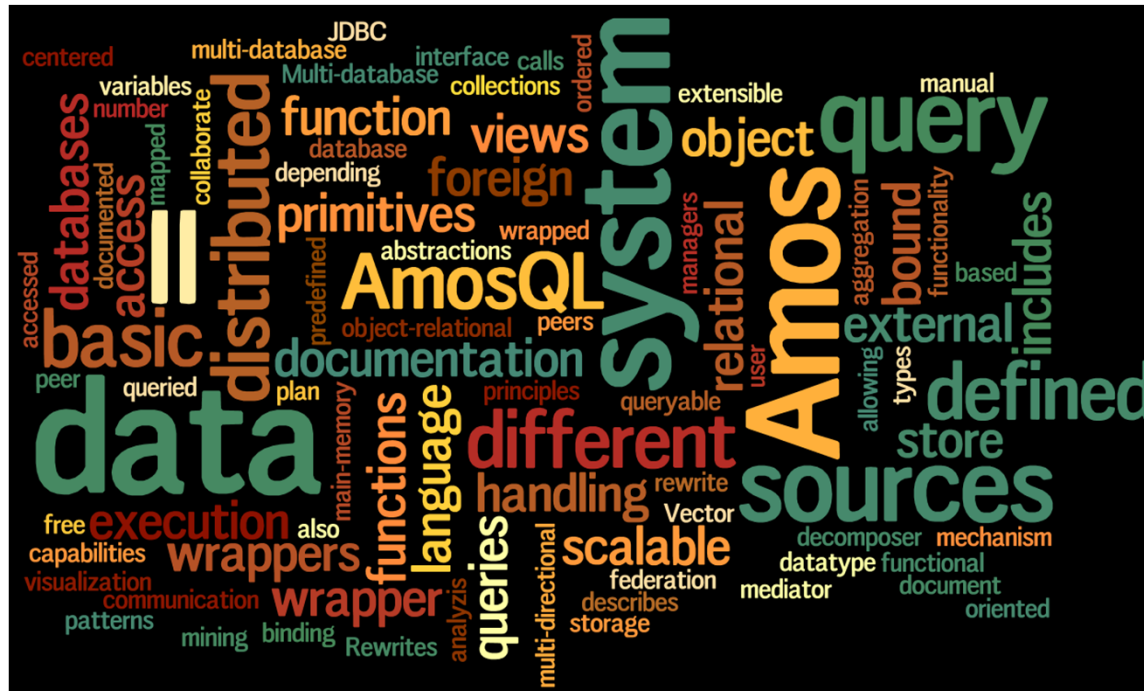
The continuation of Mexima

Thanh Truong
UDBL

“Keep things simple”



1. **Introduction**
 2. **Background**
 3. **Methodology**
 4. **Results**
 5. **Discussion**
 6. **Conclusion**
 7. **References**
 8. **Appendix**
 9. **Figure 1**
 10. **Figure 2**
 11. **Figure 3**
 12. **Figure 4**
 13. **Figure 5**
 14. **Figure 6**
 15. **Figure 7**
 16. **Figure 8**
 17. **Figure 9**
 18. **Figure 10**
 19. **Figure 11**
 20. **Figure 12**
 21. **Figure 13**
 22. **Figure 14**
 23. **Figure 15**
 24. **Figure 16**
 25. **Figure 17**
 26. **Figure 18**
 27. **Figure 19**
 28. **Figure 20**
 29. **Figure 21**
 30. **Figure 22**
 31. **Figure 23**
 32. **Figure 24**
 33. **Figure 25**
 34. **Figure 26**
 35. **Figure 27**
 36. **Figure 28**
 37. **Figure 29**
 38. **Figure 30**
 39. **Figure 31**
 40. **Figure 32**
 41. **Figure 33**
 42. **Figure 34**
 43. **Figure 35**
 44. **Figure 36**
 45. **Figure 37**
 46. **Figure 38**
 47. **Figure 39**
 48. **Figure 40**
 49. **Figure 41**
 50. **Figure 42**
 51. **Figure 43**
 52. **Figure 44**
 53. **Figure 45**
 54. **Figure 46**
 55. **Figure 47**
 56. **Figure 48**
 57. **Figure 49**
 58. **Figure 50**
 59. **Figure 51**
 60. **Figure 52**
 61. **Figure 53**
 62. **Figure 54**
 63. **Figure 55**
 64. **Figure 56**
 65. **Figure 57**
 66. **Figure 58**
 67. **Figure 59**
 68. **Figure 60**
 69. **Figure 61**
 70. **Figure 62**
 71. **Figure 63**
 72. **Figure 64**
 73. **Figure 65**
 74. **Figure 66**
 75. **Figure 67**
 76. **Figure 68**
 77. **Figure 69**
 78. **Figure 70**
 79. **Figure 71**
 80. **Figure 72**
 81. **Figure 73**
 82. **Figure 74**
 83. **Figure 75**
 84. **Figure 76**
 85. **Figure 77**
 86. **Figure 78**
 87. **Figure 79**
 88. **Figure 80**
 89. **Figure 81**
 90. **Figure 82**
 91. **Figure 83**
 92. **Figure 84**
 93. **Figure 85**
 94. **Figure 86**
 95. **Figure 87**
 96. **Figure 88**
 97. **Figure 89**
 98. **Figure 90**
 99. **Figure 91**
 100. **Figure 92**
 101. **Figure 93**
 102. **Figure 94**
 103. **Figure 95**
 104. **Figure 96**
 105. **Figure 97**
 106. **Figure 98**
 107. **Figure 99**
 108. **Figure 100**
 109. **Figure 101**
 110. **Figure 102**
 111. **Figure 103**
 112. **Figure 104**
 113. **Figure 105**
 114. **Figure 106**
 115. **Figure 107**
 116. **Figure 108**
 117. **Figure 109**
 118. **Figure 110**
 119. **Figure 111**
 120. **Figure 112**
 121. **Figure 113**
 122. **Figure 114**
 123. **Figure 115**
 124. **Figure 116**
 125. **Figure 117**
 126. **Figure 118**
 127. **Figure 119**
 128. **Figure 120**
 129. **Figure 121**
 130. **Figure 122**
 131. **Figure 123**
 132. **Figure 124**
 133. **Figure 125**
 134. **Figure 126**
 135. **Figure 127**
 136. **Figure 128**
 137. **Figure 129**
 138. **Figure 130**
 139. **Figure 131**
 140. **Figure 132**
 141. **Figure 133**
 142. **Figure 134**
 143. **Figure 135**
 144. **Figure 136**
 145. **Figure 137**
 146. **Figure 138**
 147. **Figure 139**
 148. **Figure 140**
 149. **Figure 141**
 150. **Figure 142**
 151. **Figure 143**
 152. **Figure 144**
 153. **Figure 145**
 154. **Figure 146**
 155. **Figure 147**
 156. **Figure 148**
 157. **Figure 149**
 158. **Figure 150**
 159. **Figure 151**
 160. **Figure 152**
 161. **Figure 153**
 162. **Figure 154**
 163. **Figure 155**
 164. **Figure 156**
 165. **Figure 157**
 166. **Figure 158**
 167. **Figure 159**
 168. **Figure 160**
 169. **Figure 161**
 170. **Figure 162**
 171. **Figure 163**
 172. **Figure 164**
 173. **Figure 165**
 174. **Figure 166**
 175. **Figure 167**
 176. **Figure 168**
 177. **Figure 169**
 178. **Figure 170**
 179. **Figure 171**
 180. **Figure 172**
 181. **Figure 173**
 182. **Figure 174**
 183. **Figure 175**
 184. **Figure 176**
 185. **Figure 177**
 186. **Figure 178**
 187. **Figure 179**
 188. **Figure 180**
 189. **Figure 181**
 190. **Figure 182**
 191. **Figure 183**
 192. **Figure 184**
 193. **Figure 185**
 194. **Figure 186**
 195. **Figure 187**
 196. **Figure 188**
 197. **Figure 189**
 198. **Figure 190**
 199. **Figure 191**
 200. **Figure 192**
 201. **Figure 193**
 202. **Figure 194**
 203. **Figure 195**
 204. **Figure 196**
 205. **Figure 197**
 206. **Figure 198**
 207. **Figure 199**
 208. **Figure 200**
 209. **Figure 201**
 210. **Figure 202**
 211. **Figure 203**
 212. **Figure 204**
 213. **Figure 205**
 214. **Figure 206**
 215. **Figure 207**
 216. **Figure 208**
 217. **Figure 209**



A recap of Mexima

- A mechanism to integrate new kinds of indexes.
 - ✱ Minimize the changes of integration with
 - Index Manager
 - Transaction Manager
 - Storage Manager
 - . . .
 - ✱ Keeps the usage of index transparently



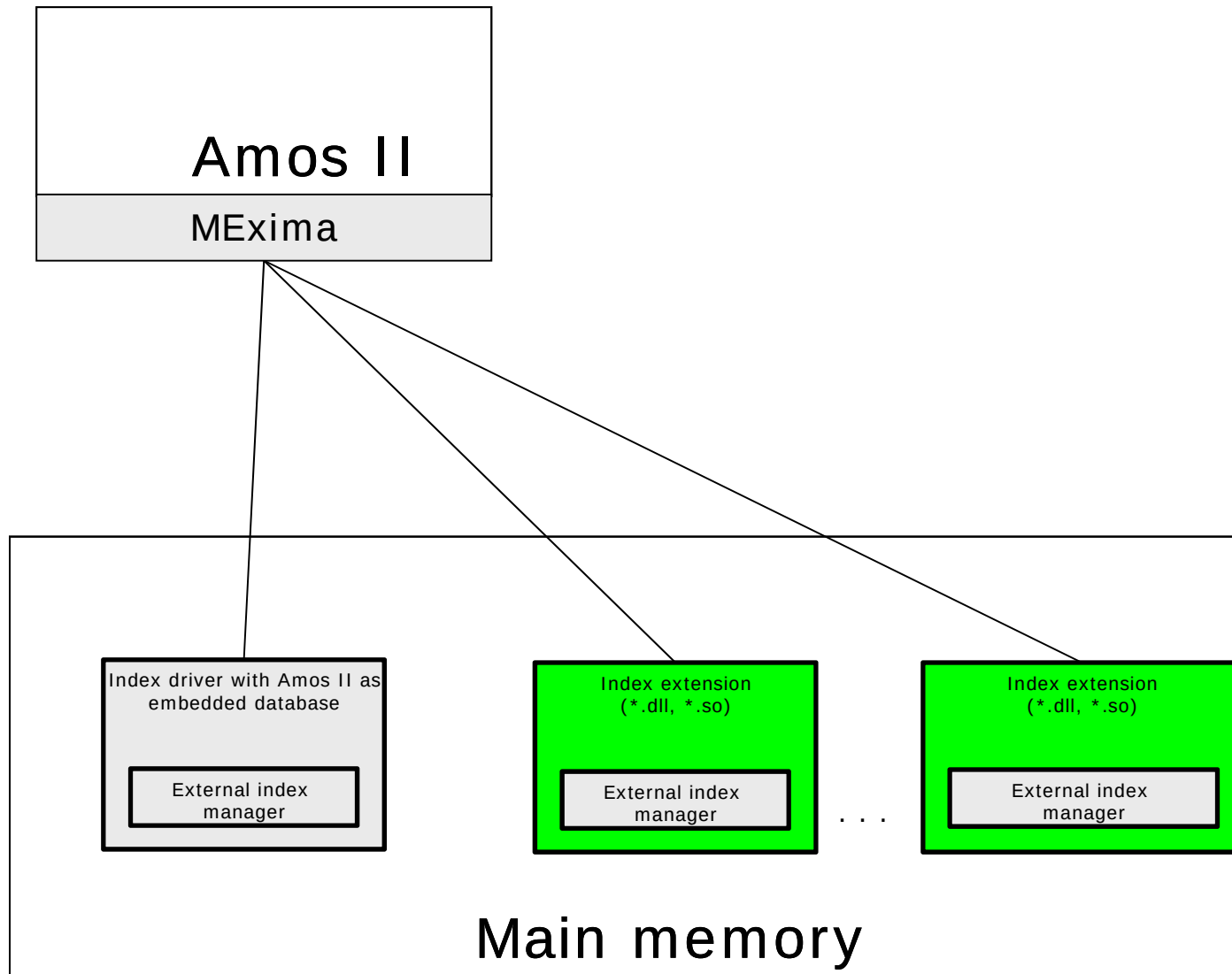
A recap of Mexima

- External index manager (Exima): an implementation of any index structure.
- Meta Exima : a framework for extending indexes.
- Index extension
 - ✱ Windows : dynamic link library (*.dll)
 - ✱ Unix : shared object (*.so)



UPPSALA
UNIVERSITET

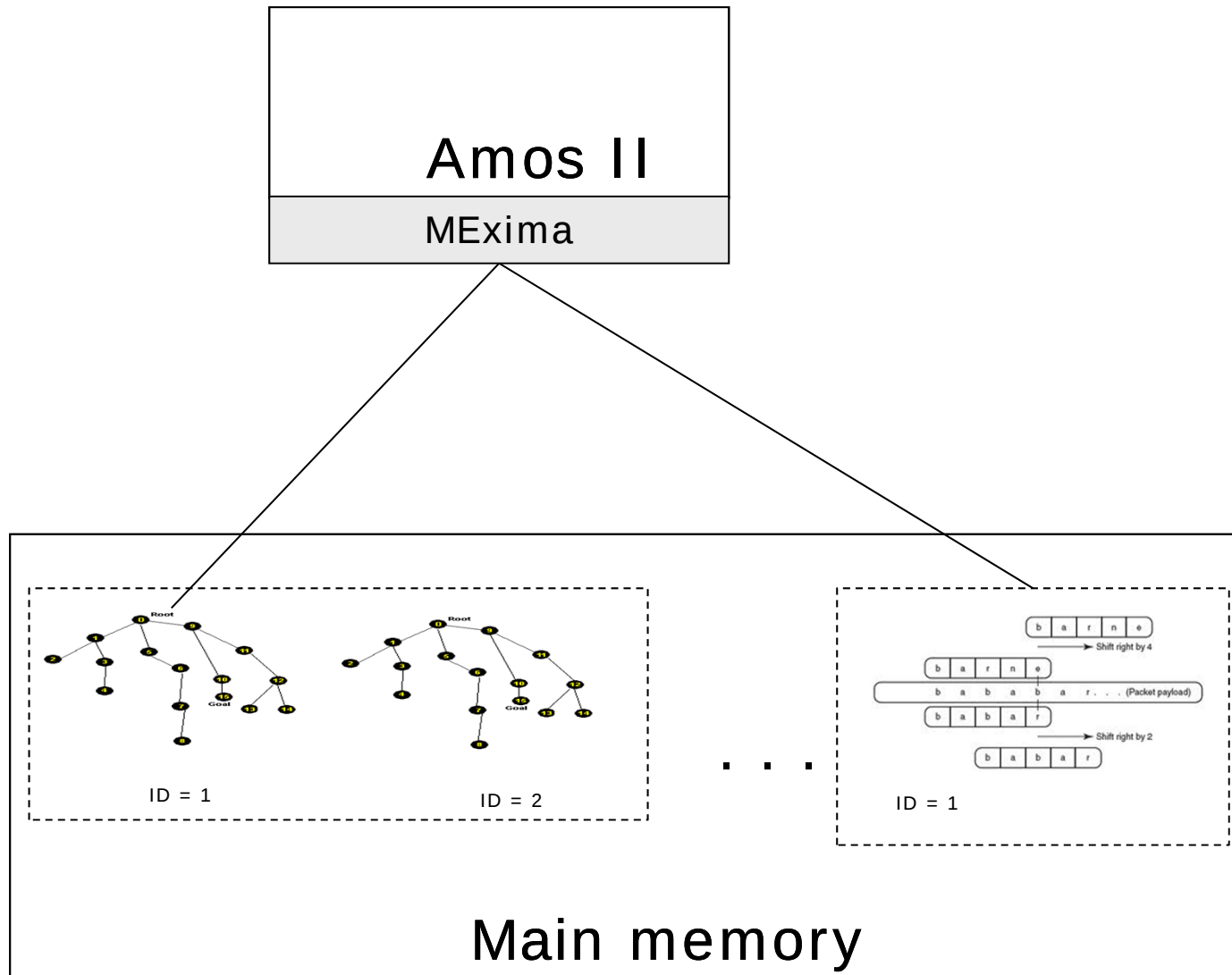
Mexima – Overview





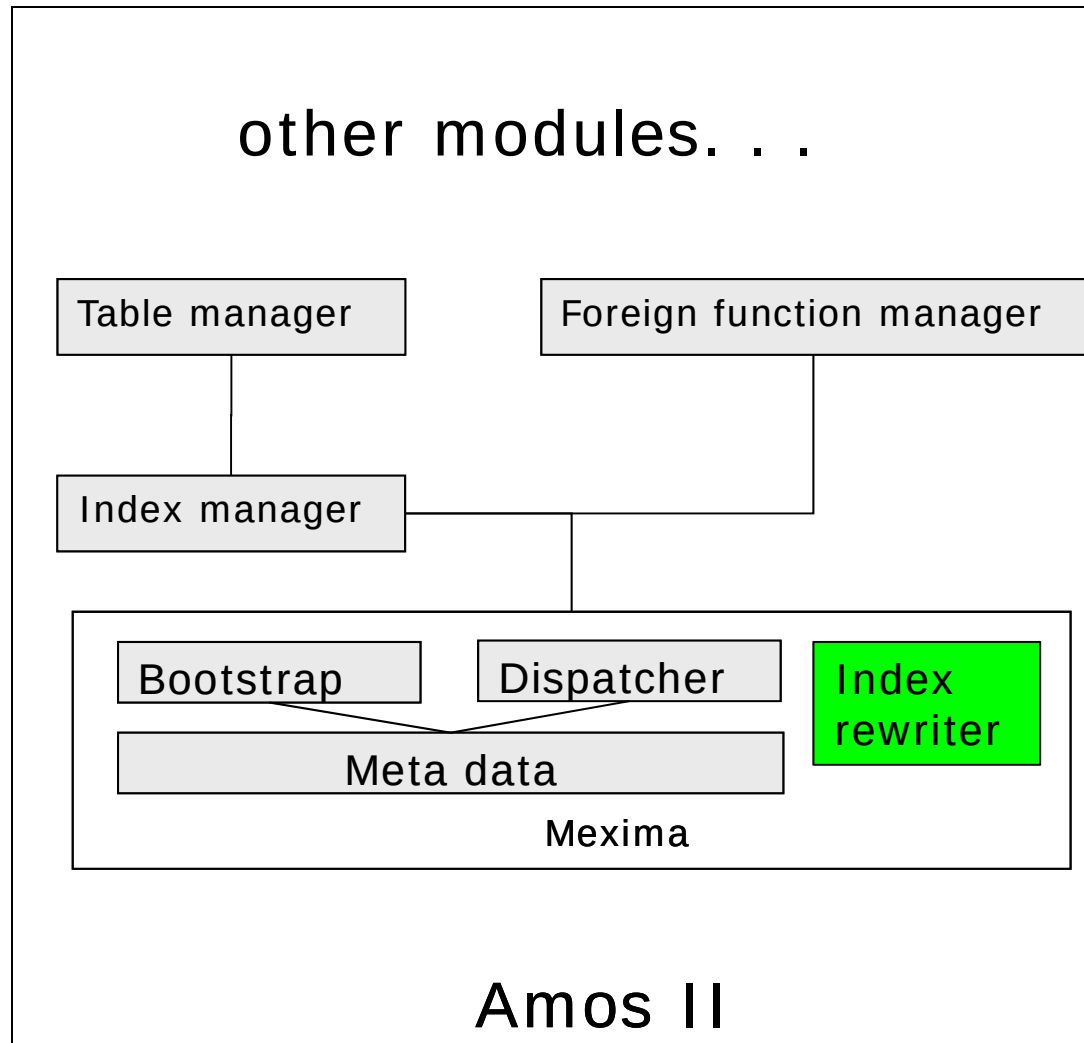
UPPSALA
UNIVERSITET

Mexima – Overview





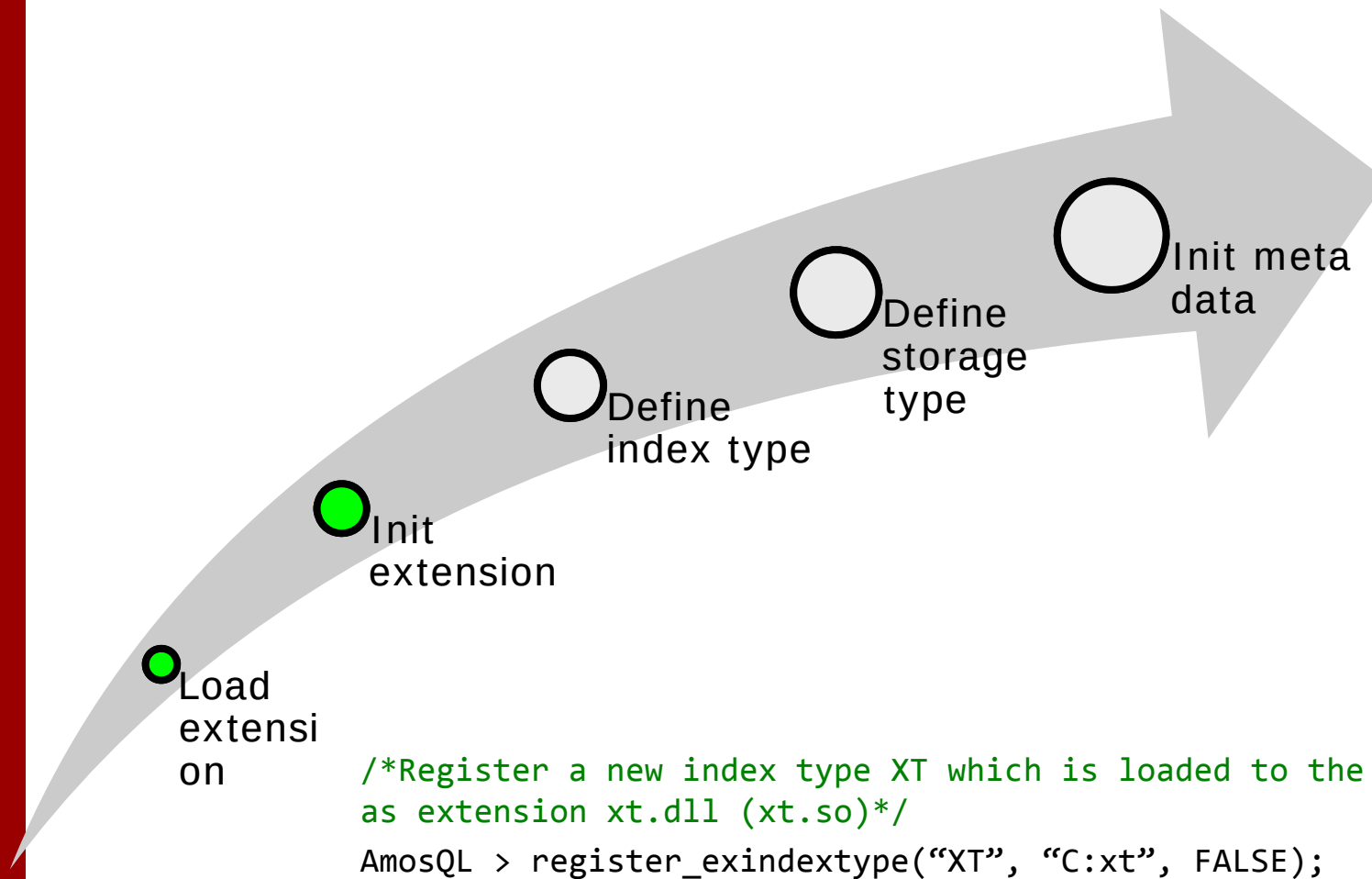
Mexima - Modules





UPPSALA
UNIVERSITET

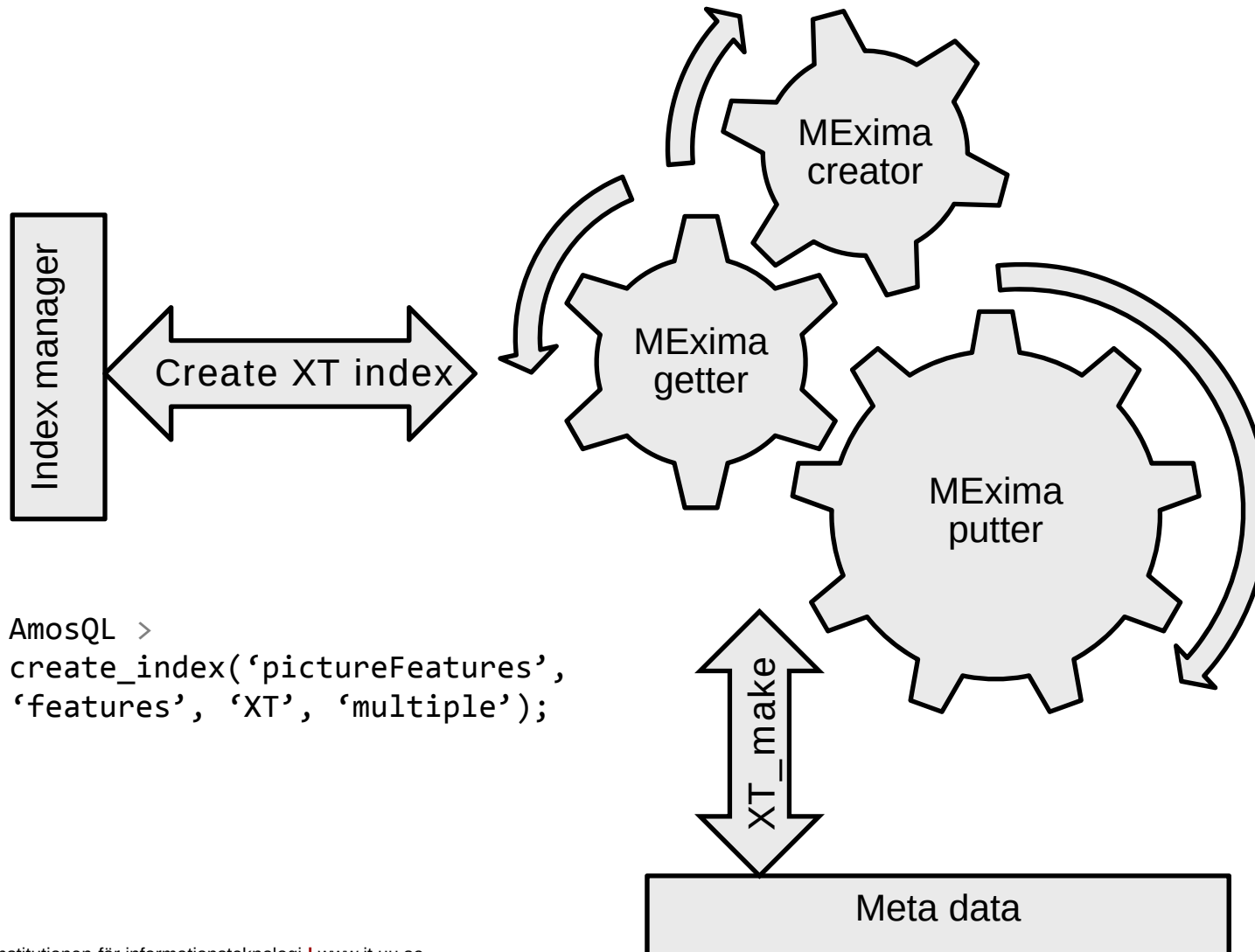
Mexima- Bootstrap





UPPSALA
UNIVERSITET

Mexima- Dispatcher

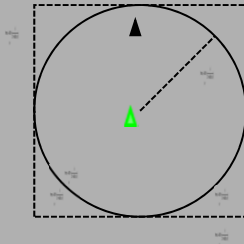




UPPSALA
UNIVERSITET

Mexima- Index rewriter

Distance search
rewrite

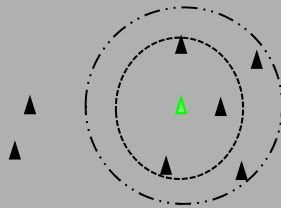


Reasoning inequality

$$\text{From } (a + 0.045) * \left(\frac{b}{2c + \text{Euclid}(x, x_0)} \right) > e$$

$$\Rightarrow \text{Euclid}(x, x_0) < (a + 0.045) * \left(\frac{b}{e} \right) - 2c$$

Index
rewriter



KNN rewrite

XT	Euclid	Xtree_distance_search
KDTree	Euclid	Kdtree_distance_search

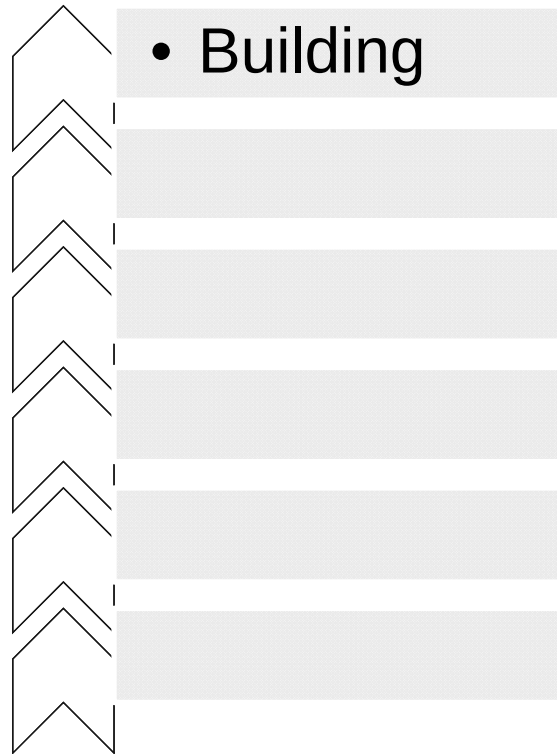
Rewrite rule matrix



UPPSALA
UNIVERSITET

Automatically building transient indexes

AFTER ROLLIN



```
/*Start Amos2 with xt.dmp*/  
amos2 xt.dmp
```



BEFORE ROLLOUT

```
/*Save to xt.dmp*/  
AmosQL > save 'xt.dmp';
```



Automatically building transient indexes

- It is shadowed by extension's manual saving/loading.
- Primary index (HASH) is created/removed in the background.
- Transient index is rebuilt when system is loaded.



Index routines

```
■ typedef oidtype (* ExinmaComputeKeyFunc) (oidtype key); /*Compute key*/

typedefint (* ExinmaCompareKeyFunc) (oidtype key1, oidtype key2);
                                     /*Compare two keys*/
typedef oidtype (* ExinmaCreateFunc)(); /*Create index*/

typedef oidtype (* ExinmaPutFunc) (int htid, oidtype key, oidtype val,
                                   ExinmaCompareKeyFunc cmpfn); /*Put key value pairs*/
typedef oidtype (* ExinmaGetFunc) (int htid, oidtype key,
                                   ExinmaCompareKeyFunc cmpfn); /*Get value given a key*/

typedef oidtype (* ExinmaDeleteFunc)(int htid, oidtype key,
                                     ExinmaCompareKeyFunc cmpfn);
                                     /*Delete index item given a
key*/
typedef oidtype (* ExinmaClearFunc) (int htid); /*Clear entire index*/

typedefint (* ExinmaMappingFunc) (char* k, char* v, void *xa);

typedef void (* ExinmaMapperFunc) (int htid, ExinmaMappingFunc mappingfn,
                                   ExinmaCompareKeyFunc cmpfn, void *xa);
/*Iterate over index structure and apply mappingfn*/
```



Index routines

```
struct ex_index_routines {
    ExinmaCreateFunc exCreateFn;
    ExinmaPutFunc exPutFn;
    ExinmaGetFunc exGetFn;
    ExinmaDeleteFunc exDeleteFn;
    ExinmaClearFunc exClearFn;
    ExinmaMapperFunc exMapperFn;
    ExinmaComputeKeyFunc exComputeKeyFn;
    ExinmaCompareKeyFunc exCompareKeyFn;
    ExinmaSaveFunc exSaveFn;
    ExinmaLoadFunc exLoadFn;
    int onLisp;    // TODO make these index routines available on Lisp
    int onAmosql; // TODO make these index routines available on AmosQL
};

/* Define a new index type and its routines*/
EXTERN void define_ex_index_type(char strIndextype[],
                                struct ex_index_routines idxro);
```



Define a new index type

/ This function will be automatically invoked when the extension is loaded*/*

```
■ EXPORT void a_initialize_extension(void) {  
    /* Define a new index type and its routines */  
    struct ex_index_routines rot;  
  
    rot.exCreateFn= xt_make;           /* Mapping to index creator*/  
    rot.exPutFn = xt_put;              /* Mapping to index putter*/  
    rot.exGetFn = xt_get;              /* Mapping to index getter*/  
    rot.exDeleteFn = xt_delete;        /* Mapping to index delete*/  
    rot.exClearFn = xt_clear;          /* Mapping to index remover*/  
    rot.exMapperFn = NULL;             /* Mapping to index mapper*/  
    rot.exComputeKeyFn = NULL;         /* Mapping to compute key function*/  
    rot.exCompareKeyFn = NULL;         /* Mapping to compare key function*/  
    rot.exSaveFn = NULL;               /* Mapping to computed key function*/  
    rot.exLoadFn = NULL;  
  
    /*Define an new index type XT and its routines*/  
    define_ex_index_type("XT", rot);  
    . . .  
}
```



Add a rewrite rule

```
/* This function will be automatically invoked when the extension is loaded*/
EXPORT void a_initialize_extension(void) {
    . . .

    /* Implementation of distance search function as a foreign function*/
    a_extfunction("xt_distance_search_fn", xt_distance_search_fn);
    amosql(" create function xt_distance_search_fn(Integer, Function f, \
        Vector of Number, Number)-> Bag of (Object, Number) \
        as foreign 'xt_distance_search_fn';", FALSE);

    /* Add a rewrite rule*/
    amosql("add_index_rewrite_rule('XT', #'euclid',
        #'XT_DISTANCE_SEARCH_FN');", FALSE);
    . . .
}
```



UPPSALA
UNIVERSITET

Informationsteknologi

Demo & Discussion